How Much Energy Does a Modal Contact Row Inject? A Cross-Formulation Measurement and a Cumulative Storage Bound

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Abstract

Reduced (modal) deformable bodies resting in unilateral contact with rigid bodies are usually resolved by a velocity-level complementarity law that couples rigid and modal unknowns through one shared multiplier. Interactive solvers run that law at small, fixed local iteration budgets. We measure what this costs. Running one transcribed contact row through three fixed-budget solver formulations — position-based (XPBD), augmented-Lagrangian (AVBD), and an implicit sequential-impulse realization — over an identical 24-cell scene × relaxation × budget sweep, we find sharply formulation-dependent behaviour. Measured against the energy contact actually dissipated, the position-based host overdraws by up to 4.4×10^7 J — 1.2×10^5 times the incident rigid kinetic energy in the most starved cells — the augmented-Lagrangian host overdraws in 23 of 24 cells but by at most 15 J, and the implicit realization never overdraws at all. Pervasive but negligible and rare but catastrophic are different pathologies, and the customary ratio-to-incident-energy diagnostic does not separate them. A budget sweep shows the amplification is a truncation artifact: it decays monotonically toward a converged reference computed in the same code path, and the gap to that reference shrinks by six orders of magnitude. Because the converged regime is unaffordable at interactive budgets, we add a cumulative modal-storage bound funded by measured contact dissipation, enforced by a reservoir ledger and a projection of the realized modal state, which holds in all 72 measured cells (60 distinct configurations). We also measure what that projection costs contact validity, and find worst-case post-projection penetration of 21.6 mm — 72% of the supporting board's thickness — with corrective impulses up to 8.7× steady state.

CCS Concepts

• **Computing methodologies → Physical simulation**.

Keywords

contact, modal reduction, passivity, position-based dynamics

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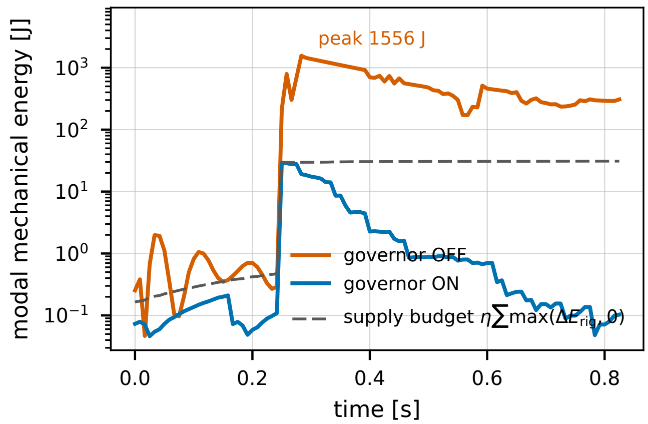

Figure 1: The same impact at the same starved budget (8×2), with and without the cumulative storage bound. The traces are identical until contact. Un-governed, modal energy runs to 1556 J in a scene whose incident rigid kinetic energy is ∼29 J; governed, it peaks at 29.3 J and stays under the running supply budget that funds it (dashed), which is measured rigid-side contact dissipation, not a tuned constant.

1 Introduction

A deformable object that is struck at one point and rings elsewhere is routinely modelled by projecting the elastic response onto a small modal basis. When such a body also rests in unilateral contact with rigid bodies, the established treatment resolves rigid and modal unknowns together through a shared contact multiplier at the velocity level [6]. The one-way alternative — driving a separate modal filter from contact events without back-reaction — is the real-time precedent [1, 2].

We take that coupled row as given. Written at position level it is one linearization step from the velocity-level law: for gap function C, $C^{n+1} \approx C^n + h\mathbf{J}\mathbf{u}^{n+1}$, which divided by h is the velocity-level complementarity condition with a C^n/h stabilization bias, carrying the same Jacobian and the same multiplier as Zheng and James [6]. *The row is not our contribution*, and ours is narrower than theirs: modes couple through normal rows only, and the position-level form takes $e = 0$ restitution.

Interactive solvers do not solve that row to convergence. They run a fixed, small number of local iterations per substep, and it is known that finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [4]. *That observation is not ours either.* What is missing is a quantitative answer for this specific case — unilateral contact-to-modal transfer — and a mitigation that fits inside an interactive budget.

Contributions.

(1) **Measurement.** One contact row, three fixed-budget solver formulations, one identical 24-cell adversarial sweep, one

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machine (§3.1); plus an iteration-budget sweep against a converged reference showing the injection is a truncation artifact (§3.2).

(2) **A cumulative storage bound.** A one-directional bound on energy entering modal storage, funded by measured rigid-side dissipation and enforced by a reservoir ledger and a projection of the realized modal state (§2); it satisfies the prescribed inequality in all 72 measured cells (60 distinct configurations).

(3) **The cost of enforcement.** Post-projection contact validity — penetration, corrective impulse, multiplier variance, momentum — measured rather than assumed (§3.3).

Relation to concurrent energy-control work. Wei et al. [4] guarantee discrete passivity for any finite inner-iteration budget by construction, using Douglas–Rachford splitting in wave coordinates whose Fejér monotonicity supplies algorithmic dissipation; their interfaces are *bilateral* couplings between subsystems, not unilateral contact, and their guarantee is structural rather than a budgeted transfer cap. You et al. [5] modify implicit Euler to track user-specified *total*-energy targets bidirectionally (dissipation or conservation) on full nonlinear elastodynamics with barrier contact. Ours is neither: a one-directional cap on how much energy contact may deposit into modal storage, funded by measured rigid-side dissipation, enforced by scaling realized modal state. For flexible bodies inside a game engine, Zobel et al. [7] solve ray-traced penalty contact quasi-statically in Unity, without a shared unilateral multiplier; reduced models built from vibration modes and modal derivatives have been used inside a position-based host as a *deformation* surrogate [3], whereas here the modal amplitudes enter as contact DOFs driven by the shared multiplier.

2 The row and the cumulative bound

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2}\dot{\mathbf{q}}^\top\dot{\mathbf{q}} + \frac{1}{2}\mathbf{q}^\top\mathbf{K}\mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C \;=\; y_c \;-\; \left(y_{\text{rest}} + \mathbf{U}_y^\top\mathbf{q}\right), \qquad C \geq 0,\ \lambda \geq 0,\ C\lambda = 0, \tag{1}$$

with $\partial C/\partial\mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law.

The invariant. We require that the energy *stored in the modal subsystem* never exceed what contact has actually dissipated on the rigid side, cumulatively over the run:

$$E^n_{\text{mod}} - E^0_{\text{mod}} \;\leq\; \eta \sum_{k\leq n} \max\!\left(\Delta E^k_{\text{rig}},\, 0\right), \qquad \eta \in (0,1], \tag{2}$$

The transfer efficiency η sets how much of the measured loss may be spent; *every experiment in this paper runs* $\eta = 1$, the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total stays inside the budget, which is what lets a legitimate ring persist across substeps.

Every term, exactly. The bound is only as meaningful as its two sides, so we give both in full. Modal storage is the mechanical energy of the reduced state, $E_{\text{mod}} = \frac{1}{2}\dot{\mathbf{q}}^\top\mathbf{M}_q\dot{\mathbf{q}} + \frac{1}{2}\mathbf{q}^\top\mathbf{K}_q\mathbf{q}$, and $E^0_{\text{mod}} = 0$: every scene starts from an undeformed, resting support, so the run's own settling transient must be funded like anything else. The supply is

$$\Delta E^k_{\text{rig}} \;=\; \left(E^{k,-}_{\text{rig}} - E^{k,+}_{\text{rig}}\right) + W_g^{\,k}, \qquad W_g^{\,k} = \sum_b m_b\,\mathbf{g}\left(\mathbf{x}_b^{k,+} - \mathbf{x}_b^{k,-}\right), \tag{3}$$

where $E_{\text{rig}} = \sum_b \left[\frac{1}{2}m_b\|\mathbf{v}_b\|^2 + \frac{1}{2}\boldsymbol{\omega}_b^\top\mathbf{I}_b\boldsymbol{\omega}_b\right]$ is *purely kinetic* — translational and rotational, summed over all dynamic bodies in the scene, not merely those in contact — and $\pm$ denote the values at the two ends of substep k. Gravity enters only through the work term W_g, which is what makes free ballistic motion register zero supply: a falling body's kinetic gain exactly equals the gravity work, so the two terms cancel and nothing is credited. Only an inelastic velocity change credits the reservoir. The endpoints are captured at the substep boundaries and tile the timeline exactly, so no rigid energy change escapes the accounting.

What is guaranteed, and what is not. Equation (2) is a *cumulative, gross-loss-funded storage ceiling on the modal subsystem.* Its supply is *source-referenced*: the right-hand side is measured rigid-side contact dissipation, never a tuned constant or a fraction of incident energy. It is not contact-port passivity, and it is not a signed per-interface transfer bound: it constrains a scalar reservoir, not a port. Two consequences of the *gross* sum $\sum \max(\Delta E_{\text{rig}}, 0)$ are worth stating plainly. It ignores substeps in which the rigid subsystem *gains* energy, so those cannot cancel earlier credits; and energy returned from the modes to the rigid bodies, then dissipated again, is counted as fresh supply the second time. We measure the size of that second effect in §4.

The enforced inequality is slightly weaker than the stated one. Reported violations use (2) exactly as printed. The implementation's own test additionally forgives one substep's largest deposit, $\max_k \eta \max(\Delta E^k_{\text{rig}}, 0)$, because modal potential energy can rise in the same substep in which the rigid body is still delivering kinetic energy — an artifact of measuring a continuous exchange at substep granularity. In these scenes that allowance is worth 7–389 J, which is immaterial where the overdraw is 10^5 J or more but decides the verdict where it is not: under it the augmented-Lagrangian host would be recorded as violating 1 of 24 cells rather than 23. We report the stricter reading throughout, for both the governed and un-governed columns, so that one inequality adjudicates both; the governed runs satisfy it with a worst margin of 1.1×10^{-13} J.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage: each substep $B \leftarrow B + \eta\max(\Delta E_{\text{rig}}, 0)$ before the contact solve, and $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0),\, 0)$ after it. If the realized modal energy would overdraw B, the realized state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \;\leftarrow\; \gamma\,(\mathbf{q}, \dot{\mathbf{q}}), \qquad \gamma \;=\; \min\left(1,\ \sqrt{\frac{E^-_{\text{mod}} + B}{E^+_{\text{mod}}}}\right), \tag{4}$$

with $E^{\mp}_{\text{mod}}$ the modal energy before and after the solve. The form is closed because E_{mod} is quadratic in the state: scaling $\mathbf{q}$ and $\dot{\mathbf{q}}$ together scales both the kinetic and the elastic term by γ^2, so γ bounds the total rather than only the velocity. Two consequences matter and are measured below. The projection is inert whenever the solve is already within budget ($\gamma = 1$, bitwise-identical trajectories); and it acts on state *after* the contact solve, so it moves the deformed

Table 1: What differs between the three hosts, and what does not. Everything below the rule is held identical; everything above it is intrinsic to the formulation. Entries are read from the implementation, with source anchors in the supplemental ledger.

	position-based	augmented-Lagr.	seq. impulse
unknown solved for	position	position + multiplier	velocity
contact treatment	compliant projection, $\tilde{\alpha} = \alpha/h^2$ (α=0: rigid)	penalty + dual update, α=0.99, β=10^5	Schur system, $\mathbf{A} = \frac{\text{cfm}}{h^2}\mathbf{I}+\mathbf{J}\mathbf{M}^{-1}\mathbf{J}^\top + \hat{\mathbf{G}}\mathbf{W}\hat{\mathbf{G}}^\top$
modal weight	explicit; per-mode compliance $1/(H_{ii}h^2-1)$	block-GS, full Newton step on $\mathbf{q}$	*implicit*, $\mathbf{W} = (\mathbf{M}+h\mathbf{D}+h^2\mathbf{K})^{-1}$
iteration	GS sweeps, gap re-evaluated each sweep	coloured primal GS + $\mathbf{q}$-block	PGS over rows
warm start	none ($\lambda\leftarrow 0$ each substep)	λ and penalty carried over	λ cache keyed per row
relaxation	0.7 on $\mathbf{q}$ and the support block	0.7 on the $\mathbf{q}$-block Newton step	*inert*: never read
h; substep	1/120 s; h/S, contacts regenerated per substep		
modal rank r	24 (shelf), 28 (ledge), 24 (table)		
modal damping	Rayleigh $\mathbf{D} = \alpha_0\mathbf{M} + \alpha_1\mathbf{K}$, $\alpha_0 \in \{2, 3\}$, α_1=10^{-5}		
stepper; η	implicit midpoint; $\eta = 1$		
machine	Apple M4, CPU only, CPython 3.12		

surface of (1) without re-solving contact — the cost quantified in §3.3.

3 Results

Every solver-behaviour number below was produced on one machine (Apple M4, CPU only, CPython 3.12); the sole exception is the device-resident timing paragraph of §3.5, which is measured on an NVIDIA RTX 4090 and reported separately. All are frozen with their generating commands and commit hashes in the supplemental ledger. Cross-machine mixing is avoided deliberately: chaotic contact stacks diverge between architectures under floating-point reassociation.

3.1 One row, three formulations, the same 24 cells

We drive an impactor into a modally-reduced support in three scenes — a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [1] — and sweep modal relaxation and the local budget (iterations × substeps) over {4×1, 8×2, 16×4, 32×8}. The reported ratio is peak modal mechanical energy over peak incident rigid kinetic energy of the impactor: an intensity diagnostic, not the enforced invariant. Table 1 lists what the three hosts do differently and what is pinned identical across them. Two entries deserve emphasis because they are the ones a controlled comparison could be accused of getting wrong. The relaxation axis is *inert* on the impulse host — its implicit modal weight needs no under-relaxation, and the attribute is never read — so its two relaxation rows are bit-identical by construction rather than by coincidence. And the hosts differ in warm start, which is intrinsic to the formulation rather than a knob we chose: the position-based host zeroes its multipliers every substep, while the other two carry them across.

Equal work, unequal outcome. A cell costs $K{\cdot}S$ row evaluations per frame, so the budget axis quadruples the work at every rung (4, 16, 64, 256) and spans ×64 end to end — steeper than the 4×1 → 32×8 labelling suggests. It also conflates two different refinements, and they are not interchangeable. Holding the budget fixed at 32 row evaluations per frame on the shelf drop and spending it entirely on iterations (K=32, S=1) gives $R = 0.300$ and satisfies (2); spending the same 32 on substeps (K=4, S=8) gives $R = 3.13$ and overdraws by 481 J. The ordering is not uniform — at 8 row evaluations substeps are marginally ahead (213.9 vs 265.7) — but iterations pull away as the budget grows (9.58 vs 68.05 at 16), and within this ladder only the iteration axis reaches the regime where the invariant holds. Substep refinement alone does not buy convergence of the contact row.

Figure 2 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation 1.0, 4×1: peak modal energy 4.44×10^7 J against an incident 371 J). The augmented-Lagrangian host exceeds $R = 1$ in 2 of 24 — both at the starved 4×1 corner of the table scene, worst 1.70 — so it is *not* unconditionally benign by this measure either. The ordering between the two hosts, however, is scene-dependent, and it inverts: taken worst-over-scenes at 4×1 the position-based host is some five orders of magnitude more energetic (1.2×10^5 vs 1.70), but *on the table scene itself* the augmented-Lagrangian host is the worse of the two (1.18 and 1.70 against a position-based worst of 0.37, which never reaches 1 there). Neither host dominates the other across scenes.

The ratio is not the invariant. R is a severity diagnostic: the incident kinetic energy of one impactor is not the supply (2) budgets against. We therefore measure (2) directly, un-governed, on all three hosts, by running the reservoir accounting live while forcing $\gamma \equiv 1$ so the projection is inert — every state write in the enforcement path is guarded by $\gamma < 1$, so the trajectory is bit-identical to an un-governed run (the ratios above reproduce the independent sweep exactly). We report the signed margin $\max_n \left[E^n_{\text{mod}} - E^0_{\text{mod}} - \eta \sum_{k \le n} \max(\Delta E^k_{\text{rig}}, 0)\right]$ in joules; positive violates. The two metrics agree on the position-based host (8/24 either way) and on the implicit one (0/24, margin at worst -5.7×10^{-4} J), but they part company on the augmented-Lagrangian host, which violates (2) in 23 of 24 cells while R flags only 2. The magnitudes are what separate the two failures: the position-based host overdraws by up to 4.4×10^7 J, the augmented-Lagrangian host by at most 15.1 J, and in 21 of its 23 cells by under 0.15 J. Pervasive but negligible overdraw and rare but catastrophic overdraw are different pathologies, and only the invariant distinguishes them.

With enforcement enabled, all 72 cells satisfy (2) — 60 of them distinct configurations, since the implicit host does not consume the relaxation axis and its 12 relaxation pairs are bit-identical by construction (Fig. 2). The worst realized ratio falls to 1.22 (XPBD) and 0.87 (AVBD), and the projection never activates in any impulse cell. At the well-resolved budgets (16×4, 32×8) the projection fires

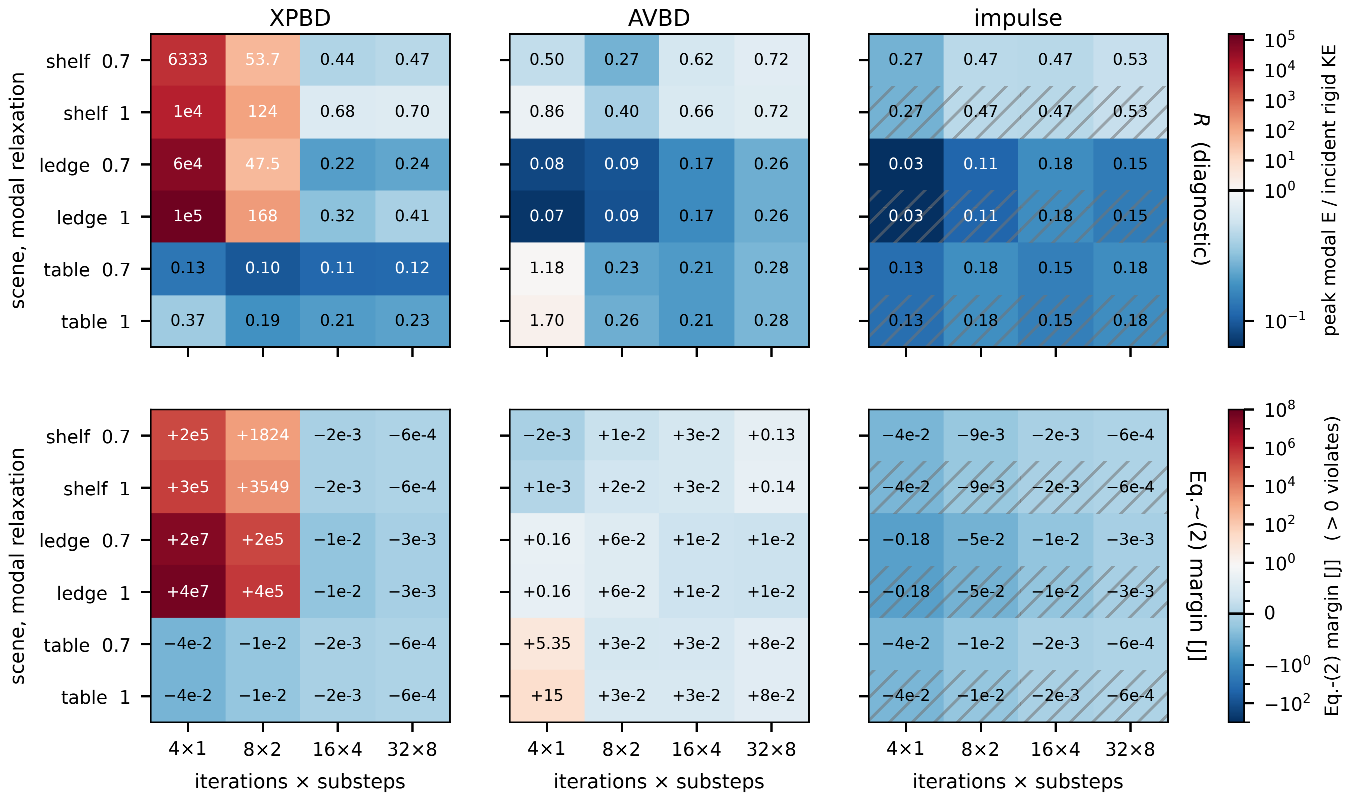

Figure 2: The same contact row in three fixed-budget solver formulations over an identical 24-cell sweep (3 scenes $\times$ modal relaxation $\{0.7, 1.0\}$ $\times$ budget), governor OFF. Top: the ratio R = peak modal energy / peak incident rigid kinetic energy, log scale diverging at 1 — a severity diagnostic. XPBD exceeds 1 in 8/24 cells, AVBD in 2/24, the implicit realization in 0/24. Bottom: the invariant itself — the signed margin of (2) in joules, symmetric-log, diverging at 0; positive violates. The rows disagree on AVBD, which violates in 23/24 cells that R mostly reports as benign, though by at most 15 J against XPBD's 4.4×10^7 J. We report joules rather than a second ratio because every candidate denominator misleads: dividing by the supply accrued so far explodes in the opening substeps, and dividing by the run total hides violations that peak early. Hatched rows on the right panels mark the relaxation axis, which that formulation does not consume (its implicit modal weight requires no under-relaxation); those rows are bit-identical by construction and are shown for grid symmetry, not as an independent measurement.

zero times on the position-based host at relaxation 0.7: it is inert exactly where the solve is already within budget.

3.2 The amplification is a truncation artifact

Figure 3 isolates the local iteration budget. The position-based host falls monotonically from 2.96×10^4 at K=1 to 0.300 at K=32, approaching the converged reference 0.2735; the gap |XPBD − reference| shrinks from 2.96×10^4 to 2.7×10^{-2}. The implicit realization moves only in the fourth decimal across $K = 2 \ldots 500$ (spread 4.9×10^{-4}).

That the energy ratio falls with K does not by itself show the *row* is converging — it could be one scalar coincidentally shrinking. Measuring the constraint directly settles it: the end-of-substep complementarity residual $\| \min(C, \lambda) \|_\infty$ on the same sweep falls monotonically from 2.79×10^{-2} at K=1 to 3.56×10^{-6} by K=64, where it bottoms out (K=128 moves it in the seventh significant figure). The energy crossing at $K \approx 24$ coincides with the residual passing $\sim 3 \times 10^{-5}$; the amplification disappears exactly as the complementarity conditions start to hold. We report this for the position-based host only: the augmented-Lagrangian multiplier is not the same object, and the implicit realization is already converged at K=2, so neither yields a comparable curve.

This is the central diagnostic result, and it cuts both ways. It confirms that the amplification is created by truncation and cured by convergence — but it equally says that iterating harder, or adopting the implicit formulation, would remove the problem outright. The bound of §2 is therefore not offered as a substitute for convergence. It is offered for the regime where neither escape is available: interactive position-based solvers ship budgets in the neighbourhood of 1×8 to 2×4, well below the $K \geq 24$ this scene needs, and the position-based formulation is what is deployed. We note explicitly that our implicit backend is a reference implementation whose cost we do not claim to be representative of that formulation class.

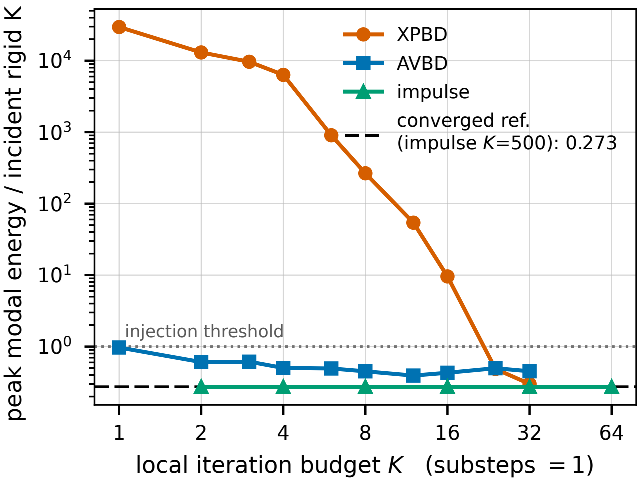

Figure 3: Energy ratio versus local iteration budget K on a deterministic shelf drop, substeps pinned at 1 so K is the only variable, governor OFF. The position-based host is monotone non-increasing in K and decays toward the converged reference, crossing the incident-energy threshold between K=16 and K=24; the gap to that reference shrinks by six orders of magnitude. The reference is the implicit realization at K=500 — the same code path run to convergence, not a different model. That realization is already converged at K=2. The AVBD curve is shown for completeness only: its denominator is itself budget-dependent at small K, so it must not be read as a convergence trend.

Table 2: Post-projection contact validity, position-based host at starved budgets, per clamp-active substep. Gap violation is measured immediately after the γ-scale; the corrective impulse is the next substep's normal impulse against the unclamped steady-state median. The 4×1 cells clamp on nearly every substep, so no steady state exists there (—).

scene	budget	clamped	gap viol. med / max [mm]	impulse	λ var.
shelf	4×1	107/108	1.25 / **21.6**	–	–
shelf	8×2	162/216	0.30 / 6.46	6.90×	26×
ledge	4×1	98/108	0.07 / **21.2**	1.03×	1.07×
ledge	8×2	96/216	0.07 / 4.08	8.69×	58×

3.3 What the projection costs contact

Equation (4) scales $\mathbf{q}$ after the contact solve has finished, which moves the surface in (1) without re-solving contact. Since $\gamma < 1$ shrinks the sag $\mathbf{U}_y^\top \mathbf{q}$, it lifts the support surface *into* the resting body. Table 2 reports the consequence, instrumented by runtime wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.25 mm, but the worst case reaches 21.6 mm, and the projection increases worst-case penetration by roughly 3.5–12.6× over its pre-scale value. That worst case is large in absolute terms and should be read against the supporting geometry: the shelf board is 30 mm thick and spans 0.8 m, so 21.6 mm is 72% of the board thickness and 2.7% of its span. It is a visible interpenetration, not a sub-millimetre artifact, and it occurs in the cell where the projection is most load-bearing (4×1, clamping on 107 of 108 substeps). Contact does recover: the violation is corrected in the following substep rather than accumulating, but that correction costs a normal impulse up to 8.7× the steady-state value, with multiplier variance up to 58× higher on clamp-active substeps. Net rigid linear momentum is unchanged across the projection to exactly zero in every clamp-active substep of every cell — necessarily so, since (4) writes only modal state; we verified it numerically rather than asserting it, and substep-to-substep momentum change on clamp-active substeps stays at or below the unclamped baseline.

3.4 The reduced response against a full-FEM reference

Bounding the energy is only useful if the response being bounded is the right one. On the ledge scene we compare the reduced arm against an unreduced FEM solve of the same mesh, with both arms built on the *same* discrete operator (the reduced basis is eigsh($\mathbf{K}, \mathbf{M}$) of the operator the reference integrates), so discretization bias cancels from every ratio. The reduced-to-reference peak-deflection ratio rises monotonically $0.38 \longrightarrow 0.54 \longrightarrow 0.79 \longrightarrow 0.88$ as the step refines $h = 1/120 \longrightarrow 1/960$ and plateaus near 0.9; the residual ~10% is k=24 mode truncation on a thick, high-frequency slab, not a coupling error — the ring frequency already agrees to 0.4% (78.0 vs 78.3 Hz). The far-field shape agrees as well: peak deflection sits at the pedestal antinode rather than under the impact, and the falloff along the slab tracks the reference at Spearman $\rho = 0.89$ with no fitted attenuation term. This is the pre-topple, small-displacement regime; we claim nothing beyond it.

3.5 Cost

The ledger adds 0.8–2.9 ms/step at the evaluated configuration on the CPU host. A separate device-resident path (NVIDIA RTX 4090), which carries the ledger read-only as a monitor rather than enforcing it, reaches 5.0–8.9 ms/step at 16×4 across four scenes — the two above plus a 2.5 × 1.5 m road slab under a dropped crate and the table scene. The shelf and ledge scenes meet a 120 Hz budget there with margin (1.7× and 1.3×); the road scene sits at the boundary (1.01× on the mean, with a 9.3 ms worst step); the table scene does not (0.94×). All four meet it at 16×2 or below. We make no unqualified real-time claim, and the governor is not enforced on that path.

4 Limitations

Stable is not accurate. When the projection is load-bearing it damps the modal response toward the budget, and §3.3 shows it opens contact violations of up to 21.6 mm — 72% of the supporting board's thickness — in the most starved cells. The result is a bounded, non-diverging simulation, not a faithful one; the honest use of the bound is as a safety envelope for budgets that would otherwise diverge.

Convergence is the better cure where affordable. Section 3.2 shows the pathology disappears with iteration count and does not appear at all in the implicit realization. Our timing for that realization is a CPU reference implementation and should not be read as the cost of that formulation class.

Scope of the row. Normal rows only — friction is body-local, so scraping and rolling transfer are outside this coupling — and the position-level form fixes $e = 0$.

The bound governs modal storage only. It constrains energy entering the modal subsystem, not energy created spuriously in rigid degrees of freedom. In the impulse realization we observe box–box rows rectifying a support ring into net rigid motion; the ledger is blind to this by design.

The supply is a gross sum, so it can be recycled. Because the reservoir is funded by $\sum_k \max(\Delta E^k_{\mathrm{rig}}, 0)$, energy that flows from the modes back to the rigid bodies and is subsequently dissipated is credited a second time. We bound the exposure by accumulating the opposite channel, $\sum_k \max(-\Delta E^k_{\mathrm{rig}}, 0)$, as a fraction of the supply: it is 0.4–27% (implicit) and 3–32% (position-based) across the sweep — not the $\lesssim$1% we expected, so the caveat stands on measurement rather than on assertion. On the augmented-Lagrangian host it reaches 102–118%: there the rigid subsystem's gross energy *gain* exceeds its gross loss in every cell, which is the same phenomenon as the box–box rectification above rather than a property of the bound. Where that holds, the supply figure should be read as an upper envelope on genuinely dissipated energy, and a signed or per-interface accounting — which this scalar reservoir deliberately is not — would be needed to tighten it.

Enforcement is host-side, single-machine, and single-precision-free (CPU float64). A device-resident governor and a validation of the *governed* path against a full-FEM reference remain future work.

Formulation coverage. Three formulations, three scenes, one contact regime. The augmented-Lagrangian result in particular (2/24 cells, worst 1.7×) shows that "descent-class solvers are passive here" is an empirical statement about this sweep, not a theorem.

5 Conclusion

Transcribing an established rigid–modal contact law into fixed-budget interactive solvers is not energy-safe, and how unsafe it is depends strongly on the solver formulation: we measure 1.2×10^5, 1.70, and 0.53 as the worst-case ratios of modal energy to incident rigid kinetic energy across position-based, augmented-Lagrangian, and implicit sequential-impulse hosts on the same 24 cells. The amplification is a truncation artifact that convergence removes. Where convergence is unaffordable, a cumulative storage bound funded by measured contact dissipation and enforced by state projection holds in every measured cell — at a contact-accuracy cost we quantify rather than omit.

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